

AnalystLab Africa

Data Analytics Capstone Project

**Education, Technology & Economic Growth:
A Comparative Analysis of Africa vs. the Rest of the World (2000–2023)**

Prepared by: Vivian Okoaze

Data Source: World Bank World Development Indicators (WDI)

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1. Project Title & Objective

Project Title: Education, Technology & Economic Growth: A Comparative Analysis of Africa vs. the Rest of the World

Problem Statement

Education and digital access are widely regarded as key drivers of economic development, yet the pace of progress on both fronts varies significantly across regions. This project investigates whether countries with higher levels of education and greater technology adoption experience stronger economic outcomes, with a specific focus on comparing Africa (defined as Sub-Saharan Africa) against the rest of the world.

Objective

The objective of this project is to apply the complete data analytics workflow: data acquisition, cleaning, analysis, visualization, and interpretation to real-world World Bank data to:

- Assess trends in education and technology adoption across regions between 2000 and 2023.
- Examine the statistical relationship between education, internet access, and GDP per capita.
- Compare economic growth patterns between Africa and the rest of the world.
- Communicate findings through an interactive Power BI dashboard and provide actionable recommendations for stakeholders.

2. Dataset Description

Source: World Bank World Development Indicators (WDI) <https://datatopics.worldbank.org/world-development-indicators/>

WDI is the World Bank's flagship collection of global development statistics, covering over 200 countries and territories across economic, social, environmental, and governance themes. The raw dataset (WDICSV.csv) contains indicator values for every country from 1960 to 2023 in wide format (one column per year). A supporting file, WDIcountry.csv, provides country-level metadata including Region and Income Group classifications, which was used to distinguish real countries from regional/income aggregates (e.g., "World," "Sub-Saharan Africa" as an aggregate entity, "OECD members").

Nine indicators across three themes were selected for this analysis:

Theme	Indicator	WDI Code
Education	Literacy rate, adult total (% ages 15+)	SE.ADT.LITR.ZS
Education	School enrollment, secondary (% gross)	SE.SEC.ENRR
Education	School enrollment, tertiary (% gross)	SE.TER.ENRR
Technology	Individuals using the Internet (% of population)	IT.NET.USER.ZS
Technology	Mobile cellular subscriptions (per 100 people)	IT.CEL.SETS.P2

Technology	Fixed broadband subscriptions (per 100 people)	IT.NET.BBND.P2
Economic	GDP per capita (current US\$)	NY.GDP.PCAP.CD
Economic	GDP growth (annual %)	NY.GDP.MKTP.KD.ZG
Economic	Unemployment, total (% of labor force)	SL.UEM.TOTL.ZS

Scope: 217 countries, spanning the years 2000–2023, grouped into two categories: Africa (Sub-Saharan Africa, 48 countries) and Rest of World (169 countries).

3. Data Cleaning Process

Data cleaning was performed in Python (pandas) via Google Colab, chosen over Power Query due to the size of the raw file (Approximately 400,000 rows) and to produce a documented, reusable cleaning script for the project's GitHub repository. The following steps were applied:

- Loaded WDICSV.csv and WDIcountry.csv from Google Drive.
- Filtered the dataset to the 9 selected indicator codes, reducing the working dataset from ~397,000 rows to 2,385 rows.
- Unpivoted (melted) the wide year-column format into a long format with dedicated Year and Value columns, producing 157,410 rows.
- Restricted the year range to 2000–2023 and dropped rows with missing values, reducing the dataset to 44,683 rows.
- Removed regional and income-group aggregates (e.g., "World," "Sub-Saharan Africa" as an aggregate, "Arab World") by retaining only rows whose Country Code had a non-null Region value in WDIcountry.csv, leaving 35,690 rows across 217 real countries.
- Checked for and confirmed zero duplicate Country–Indicator–Year combinations.
- Pivoted the cleaned long-format data into a wide fact table (one row per Country + Year, indicators as columns), producing 5,176 rows.
- Built a separate country lookup table (Country Code, Country Name, Region, Income Group) and derived a Region_Group flag classifying each country as "Africa" (Sub-Saharan Africa) or "Rest of World."
- Validated referential integrity between the two tables to confirm every country in the indicators table had a matching entry in the lookup table.
- Exported two clean CSV files: wdi_indicators_clean.csv (fact table) and country_lookup.csv (dimension table) for use in Power BI.

Limitation: "Africa" was defined strictly as Sub-Saharan Africa per WDI's regional classification. North African countries (e.g., Egypt, Morocco, Algeria) are classified by WDI under "Middle East & North Africa" and are therefore grouped under "Rest of World" in this analysis. This is a deliberate scoping decision, not a data error, and should be considered when interpreting regional comparisons.

4. Analysis Methodology

The two clean tables were loaded into Power BI Desktop and connected via a one-to-many relationship on Country Code (star schema: wdi_indicators_clean as the fact table, country_lookup as the dimension table). DAX measures were created for average values across all nine indicators, alongside two custom Pearson correlation coefficient measures.

Analytical techniques applied:

- Descriptive statistics: regional and global averages for each indicator (KPI cards).
- Trend analysis: year-over-year comparison of internet adoption and secondary enrollment, Africa vs. Rest of World (2000–2023).
- Correlation analysis: Pearson correlation coefficients between (a) internet users % and GDP per capita, and (b) secondary enrollment % and GDP per capita.
- Predictive analytics: Power BI's Key Influencers visual, used to identify which indicators most strongly predict higher GDP per capita.
- Comparative analysis: GDP growth rate and GDP per capita compared directly between Africa and the rest of the world.

5. Key Findings

- Global average GDP per capita across the 217 countries analyzed is approximately \$16,074, with North America (\$65K) and Europe & Central Asia (\$31K) far ahead of Sub-Saharan Africa (~\$2K).
- Global average internet adoption stood at 38.7% as of the latest year in the dataset, but adoption in Sub-Saharan Africa (14%) trails far behind North America (77%) and the global average.
- Secondary school enrollment is near-universal in the rest of the world (~100%) but remains meaningfully lower in Africa, and the gap has persisted rather than closed over the 2000–2023 period.
- None of the top 10 countries by tertiary enrollment are in Africa; tertiary access remains concentrated in high-income economies.
- Internet adoption shows a moderate positive correlation with GDP per capita ($r = 0.51$), notably stronger than the correlation between secondary education and GDP per capita ($r = 0.21$).
- Power BI's Key Influencers analysis identified mobile subscription rates and internet usage as the two strongest, most reliable drivers of higher GDP per capita among the indicators tested.
- Africa's average annual GDP growth rate (4.05%) is higher than the rest of the world's (3.13%), yet this faster growth has not been sufficient to close the underlying gap in GDP per capita, since Africa is growing from a much smaller economic base.

6. Insights

- A clear digital and economic divide persists: Sub-Saharan Africa remains below every other world region in both internet access and GDP per capita, despite representing 22% of the countries analyzed.
- Technology adoption appears more closely tied to economic prosperity than secondary education alone ($r = 0.51$ vs. $r = 0.21$), suggesting that closing the digital gap may offer a comparatively faster

route to economic gains than education access improvements though both matters and are likely mutually reinforcing.

- Higher GDP growth rates in Africa indicate genuine economic momentum, but growth rate alone is an incomplete measure of progress: without a stronger starting base, high growth rates will take decades to meaningfully close the per-capita income gap.
- Tertiary education remains a bottleneck specific to lower-income regions; while secondary education has become widely accessible worldwide, university-level access is still effectively a high-income-country phenomenon.

7. Recommendations

- Prioritize investment in digital infrastructure (mobile and broadband access) across Sub-Saharan Africa, given its stronger observed association with GDP per capita relative to other indicators tested.
- Pair technology access investment with targeted tertiary education funding, since secondary enrollment gains have not yet translated into comparable tertiary enrollment growth in the region.
- Policymakers and development agencies should track GDP per capita alongside GDP growth rate, since growth rate alone can overstate progress in economies starting from a low base.
- Future analysis should incorporate additional indicators (e.g., electricity access, foreign direct investment, trade openness) to test whether the internet–GDP relationship holds after controlling for other infrastructure factors, since correlation does not establish causation.

8. Conclusion

This project applied an end-to-end data analytics workflow to World Bank development data, cleaning and modeling over 35,000 records into an interactive three-page Power BI dashboard covering education, technology, and economic indicators for 217 countries between 2000 and 2023. The analysis confirms a persistent economic and digital divide between Sub-Saharan Africa and the rest of the world and finds that technology adoption is more strongly associated with economic prosperity than education access alone in this dataset. While Africa's GDP growth rate outpaces the global average, the region's low starting base means growth alone will not be sufficient to close the gap without sustained investment in both digital infrastructure and higher education. These findings, along with the supporting dashboard, are intended to give stakeholders a clear, evidence-based starting point for prioritizing development investment.